SAN IGNACIO DE LOYOLA SCHOOL

Business Proposals with Social Responsibility. Final product/service responding to a Sustainable Development Goal



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Executive Summary

ReFloor is a student business project based on a simple idea: plastic waste can be transformed into a useful product. The company plans to produce and sell outdoor floor tiles made mainly from recycled plastic bottle caps and washed sand. The project combines a practical construction product with the environmental goal of reducing plastic waste.

The initial financial and production model uses a 25 × 25 cm tile. ReFloor will collect, sort, clean, shred, process, mold, cool, inspect, and store the material before delivery. Because the product is still at the project and prototype stage, performance claims such as long-term durability or resistance must be confirmed through testing before they are used as guarantees in advertising.

The target market is concentrated mainly in Asunción and the Central Department. Our Microsoft Forms survey received 66 responses. Durability was the most selected product characteristic, and gardens and patios were the most popular intended uses. The survey also showed that most respondents were not familiar with recycled-plastic tiles, which means ReFloor must clearly explain the product and prove its quality.

The final financial model assumes a selling price of Gs. 18,000 per 25 × 25 cm tile, initial monthly sales of 4,800 units, and 2% monthly sales growth. The projected first-year sales are 64,379 tiles, equal to approximately 4,023.69 m² and Gs. 1,158,822,000 in revenue. The initial capital requirement is approximately Gs. 227,298,533, including Gs. 134,000,000 in fixed assets, Gs. 4,500,000 for S.R.L. formation, and Gs. 88,798,533 in working capital.

ReFloor is planned as a Sociedad de Responsabilidad Limitada (S.R.L.) with four founding members: Arturo Temperley, Tiago Ramos, Franco Figueredo, and Rafael Benítez. The exact ownership percentages and capital quotas will be established in the final legal agreement and are not invented in this report. The project aims to develop a reliable product, build customer trust, create an organized production process, and grow through responsible use of recycled materials.

Resumen Ejecutivo

ReFloor es un proyecto empresarial estudiantil basado en una idea sencilla: los residuos plásticos pueden transformarse en un producto útil. La empresa planea producir y vender baldosas para exteriores fabricadas principalmente con tapitas de plástico reciclado y arena lavada. El proyecto combina un producto práctico para la construcción con el objetivo ambiental de reducir los residuos plásticos.

El modelo inicial de producción y finanzas utiliza una baldosa de 25 × 25 cm. El proceso incluye recolección, clasificación, limpieza, triturado, procesamiento, moldeo, enfriamiento, control de calidad y almacenamiento. Como el producto todavía se encuentra en etapa de proyecto y prototipo, las afirmaciones sobre durabilidad o resistencia a largo plazo deberán comprobarse mediante pruebas antes de utilizarse como garantía comercial.

El mercado objetivo se concentra principalmente en Asunción y el Departamento Central. La encuesta realizada con Microsoft Forms obtuvo 66 respuestas. La durabilidad fue la característica más seleccionada, mientras que los jardines y patios fueron los usos preferidos. La mayoría de los encuestados no conocía previamente las baldosas de plástico reciclado, por lo que ReFloor deberá explicar claramente el producto y demostrar su calidad.

El modelo financiero final utiliza un precio de Gs. 18.000 por baldosa de 25 × 25 cm, ventas iniciales de 4.800 unidades mensuales y un crecimiento mensual proyectado del 2%. Para el primer año se proyectan 64.379 baldosas vendidas, equivalentes a aproximadamente 4.023,69 m² y Gs. 1.158.822.000 de ingresos. El capital inicial requerido es de aproximadamente Gs. 227.298.533, incluyendo Gs. 134.000.000 en activos fijos, Gs. 4.500.000 para la constitución de la S.R.L. y Gs. 88.798.533 de capital de trabajo.

ReFloor funcionará como una Sociedad de Responsabilidad Limitada (S.R.L.) con cuatro socios fundadores: Arturo Temperley, Tiago Ramos, Franco Figueredo y Rafael Benítez. Los porcentajes de participación y las cuotas de capital se definirán en el acuerdo legal definitivo y no se inventan en este informe. El objetivo es desarrollar un producto confiable, generar confianza en los clientes, organizar una producción eficiente y crecer mediante el uso responsable de materiales reciclados.

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Company Description Worksheet

Item	ReFloor
Business Name	ReFloor S.R.L.
Mission Statement	To create practical and high-quality recycled-plastic tiles while reducing plastic waste through responsible production and sustainable innovation.
Philosophy / Values	Innovation, sustainability, quality, responsibility, teamwork, and continuous improvement.
Vision	To become a trusted Paraguayan brand for recycled-material flooring and, over time, expand into additional products made from recovered materials.
Goals & Milestones	Short term: complete prototype testing, standardize production, launch locally, build social-media presence, and reach stable monthly sales. Corporate Social Responsibility SMART goal: by the end of ReFloor's first year of operation, donate and install enough recycled-plastic tiles to renovate at least one complete room or recreational area for a local non-profit organization, such as Fundación Casa Cuna "Dr. Carlos Santiviago." Long term: increase production capacity, expand sales across Paraguay, develop additional recycled-material products, and evaluate regional expansion.
Target Market	Homeowners, construction-related businesses, landscaping companies, hardware stores, schools, and organizations in Asunción and Central that need practical outdoor flooring and value sustainability.
Industry / Competitors	Sustainable construction materials and flooring. Main researched competitors: ECOCUR (direct recycled-flooring competitor), Casa Central, and Emporio de la Construcción (indirect conventional-flooring competitors).

Item	ReFloor
Legal Structure / Ownership	ReFloor will operate as a Sociedad de Responsabilidad Limitada (S.R.L.) in Paraguay. The four founding members are Arturo Temperley, Tiago Ramos, Franco Figueredo, and Rafael Benítez. Exact ownership percentages and capital quotas remain subject to the final legal agreement.

Products & Services Description

Item	Description
Product / Service Idea	Outdoor floor tiles made mainly from recycled plastic bottle caps and washed sand. The financial model uses a 25 × 25 cm initial tile.
Special Benefits	Gives discarded plastic a second use, supports a circular-economy approach, and offers a practical flooring alternative for patios, gardens, walkways, and similar exterior spaces.
Unique Features	Use of recycled plastic as a main material, potential for different colors/patterns, local production, and a product concept directly linked to sustainability.
Limits and Liabilities	The product is intended mainly for non-structural outdoor flooring. Long-term durability, load resistance, slip performance, and other technical claims must be confirmed by testing before commercial guarantees are made.
Production and Delivery	Plastic is collected, sorted, washed, dried, shredded, combined with approved materials, processed under controlled heat, molded, cooled, finished, inspected, stored, and delivered.
Suppliers	Recycled plastic will be obtained through collection/donation systems. Washed sand, pigments/additives, packaging, and other supplies will be purchased from local suppliers.
Intellectual Property / Special Permits	ReFloor can register its name and logo through DINAPI and must complete the business, tax, municipal, labor, environmental, and workplace-safety requirements that apply before operations begin.

Note: *The survey showed stronger interest in 40 × 40 cm and 60 × 60 cm tiles, but the final accounting and production model uses 25 × 25 cm as the standardized initial size. Larger formats can be evaluated later after prototype and machinery testing.*

Product & Prototype Visual Evidence



Figure 1. Concept illustration of the planned 25 × 25 cm ReFloor tile.



Figure 2. Illustration of ReFloor tiles in an outdoor patio/garden setting.

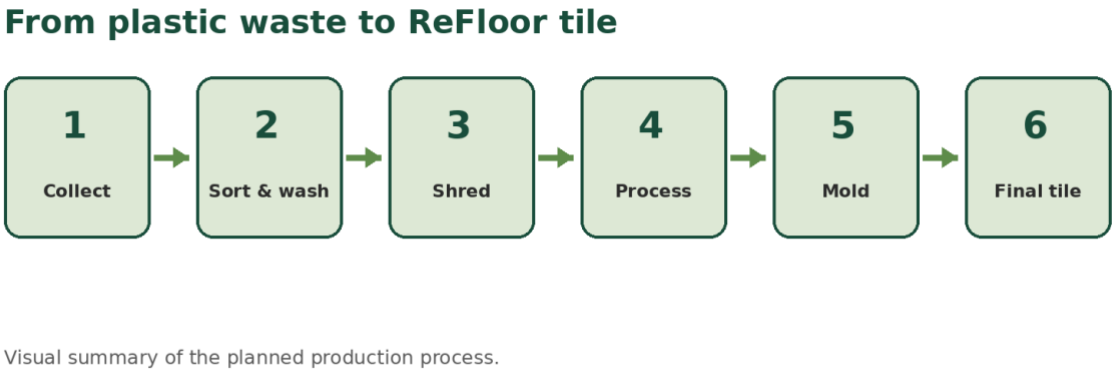


Figure 3. Illustrated process from collected plastic to the finished tile.

Note: These are product/prototype illustrations for the assignment. They show the intended appearance, use, and production concept; they are not presented as laboratory test evidence or photographs of a final commercial product.

Circular Economy

ReFloor applies circular-economy principles by using discarded plastic as a production input instead of treating it as waste. The basic cycle is: plastic collection → ReFloor production → tile use → recovery of clean production offcuts and, in the future, collection of damaged or end-of-life tiles → new recycled products. This

approach can extend the useful life of materials and reduce the amount of plastic sent to disposal sites.

During production, clean reusable offcuts should be separated and returned to future batches when technically safe. A future take-back program may also be evaluated after the company confirms that used tiles can be reprocessed without reducing product quality or creating safety problems.

Corporate Social Responsibility (CSR)

ReFloor is committed to supporting the community through its recycled products. As part of its Corporate Social Responsibility program, the company plans to collaborate with non-profit organizations by donating recycled plastic tiles to improve their facilities. One possible partner is Fundación Casa Cuna “Dr. Carlos Santiviago,” an organization that supports babies, toddlers, and children whose families need assistance with early childhood care and education.

SMART Goal

By the end of ReFloor’s first year of operation, the company aims to donate and install enough recycled plastic tiles to renovate at least one complete room or recreational area for a local non-profit organization, such as Fundación Casa Cuna “Dr. Carlos Santiviago.” This goal is specific and measurable because it establishes one completed donation project, achievable and relevant to ReFloor’s product and social-responsibility purpose, and time-bound within the company’s first year of operation.

Marketing Plan

Market Research

The ReFloor survey was conducted using Microsoft Forms and collected 66 responses. Around 51% of respondents were between 41 and 60 years old and 34% were between 26 and 40. Approximately 76% lived in Asunción and 21% in the Central Department, which supports an initial geographic focus on these two areas.

Durability was the strongest product priority, selected by 51 respondents. Ease of cleaning and design were also important. Most respondents had not previously heard of recycled-plastic tiles: 58 said they were unfamiliar with the product, while only 8 already knew about it. However, almost all respondents were either willing or possibly willing to consider using the product, showing that the main challenge is building trust rather than overcoming strong rejection.

Gardens were the most popular intended use with 36 selections, followed by patios with 28. Sidewalks and garages each received 14 selections. The survey therefore supports positioning ReFloor mainly as an outdoor flooring product at the beginning.

The most common acceptable price ranges in the survey were Gs. 20,000–30,000 and Gs. 30,000–60,000 per tile. The final financial model uses Gs. 18,000 for the 25 × 25 cm tile. This is slightly below those most common survey bands, but the price must still be validated against real production costs, customer response, and comparable flooring alternatives after prototype testing.

Barriers to Entry

- Low brand recognition because ReFloor is a new project.
- Need to prove durability, consistency, safety, and quality before customers fully trust the product.
- Competition from established flooring companies and from ECOCUR, which already sells recycled flooring in Paraguay.
- Dependence on stable collection of suitable plastic and on reliable supply of other materials.
- Need for machinery, quality-control procedures, permits, safety systems, and working capital.

Threats and Opportunities

Opportunities	Threats
Growing interest in sustainable products and recycling.	Established competitors with larger catalogs and stronger market recognition.
Potential partnerships with schools, businesses, hardware stores, contractors, and recycling initiatives.	Customers may doubt the long-term performance of recycled-plastic tiles.
Strong visual story for social media: turning waste into a finished product.	Changes in material, energy, transport, or maintenance costs.
Possibility of future sizes, colors, and recycled-material products.	Equipment problems or inconsistent raw materials may reduce production quality.

SWOT Analysis Worksheet

Area	Strengths	Weaknesses	Opportunities	Threats
Product Offering	Uses recycled plastic and has a distinctive sustainable concept.	New product with unproven long-term performance.	Growing interest in green products.	Traditional and recycled-flooring competitors.
Brand / Marketing	Clear environmental identity and simple brand concept.	Low brand awareness at launch.	Social media can visually show the process and impact.	Competitors already have established promotion and sales channels.

Area	Strengths	Weaknesses	Opportunities	Threats
Staff / HR	Clear management roles and willingness to learn.	Limited practical experience.	Training and role specialization can improve performance.	Lack of specialists could slow technical development.
Finance	Plastic input can be collected at low or zero purchase cost.	High fixed-cost structure compared with a new business.	Future partnerships or sustainability programs may support growth.	Rising operating, machinery, or supply costs.
Operations	Process can be standardized into clear production steps.	Initial production capacity and quality consistency are still developing.	Better molds, controls, and equipment can improve efficiency.	Machinery failure or unsuitable plastic can interrupt production.
Market	Strong sustainability differentiation .	The market still has low awareness of recycled-plastic tiles.	Survey participants showed interest or possible interest.	Customer preferences can change and price competition is strong.

Our strongest advantages—sustainability, a distinctive recycled-material concept, and low purchase cost for collected plastic—can help ReFloor attract attention and control part of its variable cost. However, these strengths do not eliminate the need to prove product performance, build reputation, and maintain strict quality control.

Immediate next steps are to complete prototype testing, confirm the production formula and tile dimensions, establish quality standards, validate suppliers and collection channels, and prepare a controlled market launch. Long-term goals are to increase capacity, expand distribution across Paraguay, develop additional recycled-material products, and evaluate regional opportunities.

Factors Affecting Product Supply and Demand

Supply can be affected by the availability and quality of recycled plastic, washed sand and additives; machinery capacity; production time; energy use; transport; maintenance; labor availability; and the percentage of rejected tiles. Demand can be affected by price, durability, design, customer trust, environmental value, construction activity, outdoor-renovation demand, and competitor offers.

Product/Service Features and Target Customer

Item	Summary
Features and benefits	Recycled-plastic content, outdoor use, easy-clean concept, different possible colors/patterns, and environmental value. Technical performance claims remain subject to testing.
Consumers	Homeowners and adults responsible for patios, gardens, walkways, garages, or similar improvement projects.
Businesses	Construction companies, landscaping businesses, hardware stores, schools, organizations, and other buyers needing outdoor flooring.
Geographic focus	Asunción and Central during the first stage.

Competitor Data Collection and Competitive Analysis

Note: *Earlier project worksheets used different competitors. The final report uses the most recent evidence-based competitor set: ECOCUR, Casa Central, and Emporio de la Construcción.*

ECOCUR

Field	Evidence
Type	Direct competitor: recycled flooring made from recovered tires in Paraguay.
Price evidence	Research checked August 24, 2026 reported official listings starting around Gs. 210,000 and reaching around Gs. 250,000 for some options. Product size/unit must be checked before direct price-per-m² comparison.
Key features / benefits	Recycled tire material, durability, impact absorption, anti-slip performance, sound reduction, easy maintenance, and sustainability.
Market presence	Operating in Paraguay since 2019 according to its official information; established recycled-flooring experience.
Marketing strategy	Website/catalog, sustainability messaging, product images, environmental-impact information, direct contact, colors and textures.

Casa Central

Field	Evidence
Type	Indirect competitor: ceramic and porcelain flooring retailer.
Price evidence	Examples found in the research ranged from about Gs. 29,900–42,500 per m² for some ceramic floors, with higher porcelain/premium options.

Field	Evidence
Key features / benefits	Large choice of ceramic and porcelain products, sizes, finishes, colors, and designs.
Market presence	Physical points in the Asunción metropolitan area plus online quotation/contact channels.
Marketing strategy	Website, Instagram-style visual promotion, quotations, promotions, wholesale-price messages, and product-trend content.

Emporio de la Construcción

Field	Evidence
Type	Indirect competitor: construction-material retailer with ceramic and porcelain flooring.
Price evidence	Research examples included ceramic and porcelain products across a broad range, with promotions and both lower- and higher-priced options.
Key features / benefits	Very broad catalog, many brands/sizes/finishes, physical stores, online browsing and sales.
Market presence	Stores in Ciudad del Este and Asunción, online catalog, and external sales channels.
Marketing strategy	E-commerce catalog, discounts, flash offers, email/news content, physical stores, and personalized advice.

Competitive Analysis Worksheet

Factor	ReFloor	ECOCUR	Casa Central	Emporio	Importance 1–5
Product/features	S	S	S	S	1

Factor	ReFloor	ECOCUR	Casa Central	Emporio	Importance 1–5
Price	W	W	S	S	1
Quality/materials	W	S	S	S	1
Selection/variety	W	W	S	S	3
Customer service	S	S	S	S	3
Reliability	W	S	S	S	1
Eco focus	S	S	W	W	2
Innovation/technology	S	S	W	W	3
Company reputation	W	S	Developing	S	3
Location/availability	W	W	S	S	2
Appearance/design	S	W	S	S	2
Sales method	W	S	S	S	3
Advertising/social media	W	S	S	S	4
Brand image	S	S	S	S	3

Note: *For ReFloor, price is marked as a weakness rather than automatically a strength because the final model price is Gs. 18,000 per 25 × 25 cm tile (about Gs. 288,000 per m² before installation). Direct comparisons must use the same unit, tile size, and conditions.*

Positioning & Niche

ReFloor is positioned as a sustainable and practical outdoor-flooring alternative for customers who want to reuse plastic waste through a real product. Its

strongest niche is homeowners and organizations in Asunción and Central that value environmental responsibility but still expect durability, quality, and a useful design.

ReFloor should not position itself simply as “the cheapest” option. The final financial price is not necessarily lower than conventional flooring when converted to square meters. The stronger competitive message is the combination of recycled material, local production, distinctive appearance, and circular-economy purpose, supported by evidence of product quality.

Marketing and Promotional Budget

How We Will Market Our Product

ReFloor will mainly use Instagram and TikTok to show the transformation from collected plastic to finished tile, prototype tests, product uses, colors, and installation examples. Content can be posted approximately three times per week, while paid advertising should be used selectively to reach people interested in construction, home improvement, outdoor spaces, and sustainability in Asunción and Central.

The company can also use a simple website or landing page with product information, contact details, price, photos, and ordering information. Printed flyers, business cards, samples, school/community activities, fairs, and direct contact with hardware stores or contractors can support the digital strategy.

Brand Image and Design

The ReFloor brand identity is designed to connect recycling directly with flooring. The official logo combines a stylized “R”, a leaf, and floor-tile shapes. The symbol represents recycling, environmental responsibility, and the product itself. The logo must always keep its original proportions and should not be stretched, rotated, recolored, or covered with effects.

Brand Identity and Manual

The name ReFloor combines “Re”, representing recycling and reuse, with “Floor”, representing the recycled-plastic tiles produced by the project. The brand is positioned as an eco-friendly and practical alternative for customers who want outdoor flooring while supporting the reuse of plastic waste.

The final Brand Manual standardizes the logo, colors, typography, and examples of use. Forest Green (#184D3B) is the main brand color, supported by Leaf Green (#5E8C4A), Soft Green (#DDE8D5), Natural Cream (#F4F0E6), and Charcoal (#2B2B2B). Arial Bold is used for headings and Arial Regular for body text. The same system should be used in official documents, social media, packaging, flyers, and other promotional materials.

The final manual also includes three practical mockups: product packaging, an Instagram/TikTok post, and a business card/flyer. These examples show how the same visual identity can be applied consistently in real promotional materials.

Promotional Budget

To keep the communication plan aligned with the financial model, ReFloor will use the budgeted Gs. 800,000 per month for advertising and promotion. The final financial plan does not assume an additional separate Gs. 2,500,000 startup marketing expense. Initial launch materials and first campaigns must therefore fit within the monthly advertising allocation or be approved as a specific change to the financial model.

Target Market	Main Message	Main Channels	Budget Approach
Homeowners in Asunción and Central	Durability, practical outdoor use, and recycled material.	Instagram, TikTok, referrals, samples.	Main priority within the monthly budget.
Construction companies / contractors / hardware stores	Consistent supply, product data, direct quotation, sustainability.	Direct contact, samples, business cards, website.	Low-cost direct selling; selective samples.
Schools / organizations / parks	Environmental value and visible recycling impact.	Events, presentations, direct proposals, social media.	Selective campaigns when opportunities appear.

Pricing Strategy

Name of the company ReFloor SRL

Which of the pricing strategies are you using?

Skimming	Value Based	Cost - Plus
		X

ReFloor uses a Cost-Plus Pricing strategy for its recycled plastic tiles for outdoor areas. The proposed selling price is Gs. 60,000 per m². The price was determined by considering the main production and operating costs, including washed sand, labor, rent, utilities, equipment, supplies, and other production expenses, and then adding a reasonable profit margin. ReFloor also considers the prices of other flooring and outdoor tile alternatives so the product can remain competitive. The main target market is customers in Asunción and Central who are looking for durable, practical, and eco-friendly outdoor flooring. The company is positioned as an affordable, durable, and environmentally friendly alternative made mainly from recycled plastic. A point to discuss with the teacher is whether the Gs. 60,000 per m² selling price provides a sufficient profit margin after all production and operating costs are included.

Pricing / Viability Indicator	Final Model
Selling price per tile	Gs. 18,000
Equivalent selling price per m²	Gs. 288,000
Variable cost per tile	Gs. 2,820
Contribution per tile	Gs. 15,180
Contribution margin	84.3%
Monthly fixed expenses	Gs. 44,399,267
Break-even point	2,925 tiles ≈ 183 m²/month
Initial monthly forecast	4,800 tiles = 300 m²
Year 1 projected revenue	Gs. 1,158,822,000
Projected profit after IRE	Gs. 410,506,578

Table. Pricing, expenses, contribution, break-even, and projected profitability used to demonstrate viability.

Distribution Channel Assessment

Factor	Direct Sales	Online / Social Media	Construction & Hardware Stores
Ease of Entry	High	High	Medium
Geographic Proximity	Strong in Asunción/Central	Can reach customers across Paraguay	Depends on store locations

Factor	Direct Sales	Online / Social Media	Construction & Hardware Stores
Costs	Low	Low	Medium/High due to retailer margin
Competitors' Positions	Medium competition	High online competition	High established-brand competition
Management Experience	Basic sales skills required	Social-media management required	Retail relationship management required
Staffing Capabilities	Current team can manage early orders	Few additional staff required	More coordination needed
Marketing Needs	Samples, quotations, recommendations	Content, ads, messaging	Displays, brochures, samples

12-Month Sales Forecast / Pronóstico de Venta de 12 meses

Month	Units Sold	Price (Gs./tile)	Sales (Gs.)	Equivalent Area (m²)
January	4,800	18,000	86,400,000	300.00
February	4,896	18,000	88,128,000	306.00
March	4,994	18,000	89,892,000	312.13
April	5,094	18,000	91,692,000	318.38
May	5,196	18,000	93,528,000	324.75
June	5,300	18,000	95,400,000	331.25
July	5,406	18,000	97,308,000	337.88
August	5,514	18,000	99,252,000	344.63
September	5,624	18,000	101,232,000	351.50
October	5,736	18,000	103,248,000	358.50
November	5,851	18,000	105,318,000	365.69
December	5,968	18,000	107,424,000	373.00
TOTAL	64,379	—	1,158,822,000	4,023.69

Note: *The forecast starts at 4,800 tiles per month (approximately 300 m²) and applies approximately 2% monthly growth. The forecast is a planning model, not a guarantee of actual sales.*

Operational Plan / Plan Operativo

Production / Producción

ReFloor will use a small-scale production process. Suitable plastic must be sorted and cleaned before heating. The operational plan identifies HDPE and PP as preferred recyclable thermoplastics and excludes PVC and unidentified plastics from the normal line. Washed sand and approved additives can be included according to the final tested formula.

1. Reception and sorting of plastic by type and condition.
2. Washing and complete drying.
3. Shredding into consistent flakes.
4. Batch preparation and weighing of materials.
5. Controlled heating/mixing in appropriate equipment with ventilation.
6. Molding and pressing into the 25 × 25 cm initial format.
7. Cooling and demolding.
8. Finishing and removal of defects or sharp edges.
9. Inspection, batch identification, storage, and delivery.

Quality Control / Control de calidad

Each batch should be traceable. Incoming material must be clean and suitable; the company should record source and weight, verify dryness, inspect molds and equipment, check tile dimensions and flatness, identify visible defects, and document rejected products. Finished tiles should be stored flat and protected from conditions that could deform or contaminate them.

Before launch, sample tiles should be tested for the performance characteristics that ReFloor plans to advertise, such as resistance to water exposure, impact, abrasion, and load. Defective tiles must not be sold. Customer complaints should be recorded and reviewed to identify recurring problems.

Location / Ubicación

ReFloor plans to operate from a leased workshop or light-industrial facility in Asunción, in an area where the intended activity is compatible with municipal land-use rules. The final property should provide ventilation, safe machinery space, raw-material and finished-product storage, loading access, utilities, fire protection, and a

small administration area. The financial model currently assumes Gs. 2,500,000 per month for rent, which must be validated with real quotations before launch.

Legal Environment / Entorno jurídico

The company plans to operate as an S.R.L. Before commercial operation, the project must complete the applicable legal and administrative steps, including company formation, RUC registration with DNIT, municipal requirements, employer and employee procedures with MTESS and IPS, trademark registration with DINAPI, and environmental authorization or consultation with MADES when required. The exact permits and costs should be confirmed with the competent authorities and professional advisers before operations begin.

Personnel / Personal

The financial model uses 10 employees at the assumed minimum monthly salary of Gs. 3,044,000, plus a 16.5% employer contribution and annual aguinaldo provision. An initial organization can include production supervision, quality/safety coordination, material reception and washing, shredding/preparation, press operation, finishing/warehouse work, and sales/administration support. Only trained and authorized adults should operate industrial heating, shredding, or pressing equipment.

Inventory / Inventario

ReFloor will control both raw materials and finished tiles. Plastic, sand, additives, packaging, and finished products should be stored in clean, dry, identified areas. The financial model uses a 5% safety-stock assumption. Because tiles are non-perishable, inventory control should focus on physical condition, deformation, contamination, batch traceability, and quantity rather than expiration dates.

Suppliers / Proveedores

The main paid suppliers are expected to provide washed sand, pigments/additives, packaging, maintenance inputs, and other operating supplies. Recycled plastic caps are planned to come mainly from collection and donation. The financial model assumes supplier payments of 60% in cash and 40% at 30 days.

Credit Policy / Política de crédito

The final model assumes that 70% of sales are collected in cash and 30% are collected at 30 days. Credit should only be granted to customers that meet the company's basic approval criteria, with clear invoice, payment date, and follow-up procedures. The company should avoid offering longer credit terms until cash flow and customer payment behavior are proven.

Management Worksheet and Organization Chart

Member	Main Position	Main Responsibilities	Skills / Strengths
Arturo Temperley	General Management / CEO	Coordinates the company, connects departments, follows objectives, and supports major decisions.	Leadership, communication, organization, decision-making.
Tiago Ramos	Production & Operations	Supervises production, workers, equipment use, inventory, and quality procedures.	Organization, problem-solving, teamwork, production management.
Franco Figueredo	Commercial, Marketing & Sales	Manages promotion, social media, customer contact, sales support, and brand communication.	Creativity, communication, marketing, customer relations.
Rafael Benítez	Finance & Administration	Controls budgets, accounting information, payments, HR administration, and financial planning.	Responsibility, organization, financial planning, analytical skills.

Organization Chart

Level / Area	Responsible Structure
Board of Directors	Arturo Temperley • Tiago Ramos • Franco Figueredo • Rafael Benítez

Level / Area	Responsible Structure
CEO / General Manager	Arturo Temperley
Production & Operations	Tiago Ramos
Commercial / Marketing / Sales	Franco Figueredo
Finance / Accounting / Administration / HR	Rafael Benítez
Quality & Sustainability Coordination	Operational support role; works with Production and management.
Operational Staff	Production workers, inventory/warehouse support, accounting/purchasing support, and customer-service support as needed.

Note: *Because ReFloor is a small project, one founder may cover more than one management function at the beginning. Responsibilities should remain clear even when roles are combined.*

Gastos Iniciales y Capitalización

- **Gastos de puesta en marcha**

PLAN DE COMPRAS Y MATERIA PRIMA – REFLOOR

Concepto	ENERO	FEBRERO	MARZO	ABRIL	MAYO	JUNIO	JULIO	AGOSTO	SEPTIEMBRE	OCTUBRE	NOVIEMBRE	DICIEMBRE	TOTAL
Producción requerida (baldosas)	**5,040**	**5,141**	**5,244**	5,349	5,456	5,565	5,676	5,790	5,905	6,023	6,144	6,266	67,599
Arena lavada (kg)	**10,080**	**10,282**	**10,488**	**10,698**	**10,912**	**11,130**	**11,352**	**11,580**	**11,810**	**12,046**	**12,288**	**12,532**	**135,198**
Costo arena lavada	6,048,000	6,169,200	6,292,800	6,418,800	6,547,200	6,678,000	6,811,200	6,948,000	7,086,000	7,227,600	7,372,800	7,519,200	81,118,800
Tapitas recicladas (kg)	1,764	1,799	1,835	1,872	1,910	1,948	1,987	2,027	2,067	2,108	2,150	2,193	**23,660**
Costo tapitas	-	-	-	-	-	-	-	-	-	-	-	-	-
Pigmentos / aditivos	3,528,000	3,598,700	3,670,800	3,744,300	3,819,200	3,895,500	3,973,200	4,053,000	4,133,500	4,216,100	4,300,800	4,386,200	47,319,300
Empaque / protección	2,520,000	2,570,500	2,622,000	2,674,500	2,728,000	2,782,500	2,838,000	2,895,000	2,952,500	3,011,500	3,072,000	3,133,000	33,799,500
Subtotal materiales	**12,096,000**	**12,338,400**	**12,585,600**	**12,837,600**	**13,094,400**	**13,356,000**	**13,622,400**	**13,896,000**	**14,172,000**	**14,455,200**	**14,745,600**	**15,038,400**	**162,237,600**
Merma de producción	604,800	616,920	629,280	641,880	654,720	667,800	681,120	694,800	708,600	722,760	737,280	751,920	8,111,880
TOTAL COMPRAS	**12,700,800**	**12,955,320**	**13,214,880**	**13,479,480**	**13,749,120**	**14,023,800**	**14,303,520**	**14,590,800**	**14,880,600**	**15,177,960**	**15,482,880**	**15,790,320**	**170,349,480**

- **Balance del día de apertura**

BALANCE SIMPLIFICADO				
ACTIVO	Importe			
Caja	518,568,781			
Cuentas por cobrar	32,227,200			
Inventario / stock	789,516			
Activo fijo	134,000,000			
TOTAL ACTIVO	685,585,497			
PASIVO Y PATRIMONIO				
Proveedores	6,316,128			
Capital + resultado	637,411,721			

Plan Financiero

- **Proyección de pérdidas y ganancias a 12 meses**

OSA REFLOOR 25 x 25 CM

Concepto			MARZO	ABRIL	MAYO	JUNIO	JULIO	AGOSTO	SEPTIEMBRE	OCTUBRE	NOVIEMBRE	DICIEMBRE	TOTAL
Cantidad vendida (unid.)	4,800	4,896	4,994	5,094	5,196	5,300	5,406	5,514	5,624	5,736	5,851	5,968	64,379
Precio de venta (Gs./unid.)	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000
Ventas (Gs.)	86,400,000	88,128,000	89,892,000	91,692,000	93,528,000	95,400,000	97,308,000	99,252,000	101,232,000	103,248,000	105,318,000	107,424,000	1,158,822,000
Área equivalente (m^2)	300	306	312	318	325	331	338	345	352	359	366	372	4,024
	25,920,000,000	26,967,168,000	28,057,540,500	29,192,440,500	30,373,218,000	31,601,250,000	32,877,940,500	34,204,720,500	35,583,048,000	37,014,408,000	38,513,476,125	40,069,152,000	239,194,278,000

- **Proyección de flujo de efectivo**

FLUJO DE CAJA PROYECTADO – REFLOOR S.R.L.													
	Concepto												
	Saldo inicial	-	97,258,787	127,615,608	159,468,085	192,847,179	227,783,848	264,309,053	302,453,755	342,247,400	383,721,453	426,906,371	471,843,192	-
	Aporte de capital	227,298,533	-	-	-	-	-	-	-	-	-	-	-	227,298,533
	Cobros por ventas	60,480,000	87,609,600	89,362,800	91,152,000	92,977,200	94,838,400	96,735,600	98,668,800	100,638,000	102,643,200	104,697,000	106,792,200	1,126,594,800
	TOTAL INGRESOS	287,778,533	184,868,387	216,978,408	250,620,085	285,824,379	322,622,248	361,044,653	401,122,555	442,885,400	486,364,653	531,603,371	578,635,392	1,353,893,333
	Pago a proveedores	7,620,480	12,853,512	13,111,056	13,373,640	13,641,264	13,913,928	14,191,632	14,475,888	14,764,680	15,059,016	15,360,912	15,667,344	164,033,352
	Salarios	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	30,440,000	365,280,000
	Aporte patronal	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	5,022,600	60,271,200
	Alquiler	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	30,000,000
	Servicios básicos	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	12,000,000
	Mantenimiento	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	9,600,000
	Seguridad e higiene	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	6,000,000
	Papelería y administración	300,000	300,000	300,000	300,000	300,000	300,000	300,000	300,000	300,000	300,000	300,000	300,000	3,600,000
	Honorarios profesionales	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	500,000	6,000,000
	Publicidad	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	800,000	9,600,000
	Aguinaldo (provisión)	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	2,536,667	30,440,000
	Activo fijo	134,000,000	-	-	-	-	-	-	-	-	-	-	-	134,000,000
	Constitución S.R.L.	4,500,000	-	-	-	-	-	-	-	-	-	-	-	4,500,000
	TOTAL EGRESOS	190,519,747	57,252,779	57,510,323	57,772,907	58,040,531	58,313,195	58,590,899	58,875,155	59,163,947	59,458,283	59,760,179	60,066,611	835,324,552
	SALDO FINAL	97,258,787	127,615,608	159,468,085	192,847,179	227,783,848	264,309,053	302,453,755	342,247,400	383,721,453	426,906,371	471,843,192	518,568,781	3,515,023,512

- **Balance proyectado**

BALANCE GENERAL PROYECTADO – REFLOOR S.R.L.	
ACTIVO	**Importe (Gs.)**
Caja y bancos	518,568,781
Cuentas por cobrar	32,227,200
Inventario de productos terminados	46,500
Activo fijo	134,000,000
TOTAL ACTIVO	**684,842,481**
PASIVO	**Importe (Gs.)**
Proveedores por pagar	6,316,128
TOTAL PASIVO	**6,316,128**
PATRIMONIO	**Importe (Gs.)**
Patrimonio proyectado	678,526,353
TOTAL PATRIMONIO	**678,526,353**
TOTAL PASIVO + PATRIMONIO	**684,842,481**
DIFERENCIA DE BALANCE	-

- **Cálculo del punto de equilibrio – *Break-even analysis***

Breakeven Analysis

SCORE

Concepto	Valor	Unidad	Interpretación
Precio de venta por baldosa	18,000.00	Gs./unidad	Precio de venta
Costo variable unitario	2,820.00	Gs./unidad	Costo de producción por baldosa
Margen de contribución	**15,180.00**	**Gs./unidad**	**Precio - costo variable**
Costos fijos mensuales	44,399,266.67	**Gs./mes**	Costos fijos estimados
Punto de equilibrio	2,924.85	baldosas/mes	Costos fijos ÷ margen
Ventas de equilibrio	52,647,351.78	Gs./mes	Ingresos necesarios para no tener
Área equivalente	182.80	m^2/mes	16 baldosas = 1 m^2
Margen sobre precio	84.33%	%	Margen de contribución / precio

Escenario			
Conservador	**3,000**	**54,000,000**	**1,140,733**
Base	4,800	86,400,000	28,464,733
Crecimiento	6,000	108,000,000	46,680,733
Fuerte	8,000	144,000,000	77,040,733

RESUMEN

EL PUNTO DE EQUILIBRIO ES 2.925 BALDOSAS

≈ 183 m^2 de baldosas por mes

- **Uso del capital**

Activo fijo inicial

Cant.	Descripción	Precio unitario IVA incl.	Total sin IVA	IVA	Total IVA incl.	Categoría
1	**Sillas**	**6,000,000**	**5,454,545**	**545,455**	**6,000,000**	Bienes de operación
1	**Espátulas**	**300,000**	**272,727**	**27,273**	**300,000**	Bienes de operación
1	Silla industrial	20,000,000	18,181,818	1,818,182	20,000,000	Bienes de operación
1	Calentador industrial	4,500,000	4,090,909	409,091	4,500,000	Maquinaria
1	Paletín	200,000	181,818	18,182	200,000	Herramientas
1	Mesa de trabajo	2,000,000	1,818,182	181,818	2,000,000	Bienes de operación
1	Molde para baldosas 25×25	1,000,000	909,091	90,909	1,000,000	Producción
1	Trituradora industrial	70,000,000	63,636,364	6,363,636	70,000,000	Maquinaria
1	Impresora	2,000,000	1,818,182	181,818	2,000,000	Informática
1	Laptop	3,000,000	2,727,273	272,727	3,000,000	Informática
1	**Vehículo / flete de reparto**	**25,000,000**	**22,727,273**	**2,272,727**	**25,000,000**	Logística
	TOTAL ACTIVO FIJO		**121,818,182**	**12,181,818**	**134,000,000**	

Gastos operativos fijos

Nº	Gasto	Base	Mensual	Anual	Frecuencia	Observación
1	Salarios	10 × salario mínimo	30,440,000	365,280,000	Mensual	10 empleados
2	Aporte patronal	16,5% salarios	5,022,600	60,271,200	**Mensual**	IPS patronal
3	Alquiler	Local/taller	2,500,000	30,000,000	Mensual	
4	**Servicios básicos**	Agua, luz, teléfono e internet	1,000,000	12,000,000	Mensual	
5	**Mantenimiento**	Maquinaria y herramientas	800,000	9,600,000	Mensual	
6	Seguridad e higiene	EPP y limpieza industrial	500,000	6,000,000	Mensual	
7	Papelería y administración	Oficina	300,000	3,600,000	**Mensual**	
8	Honorarios profesionales	Contabilidad / asesoría	500,000	6,000,000	Mensual	
9	**Publicidad**	Promoción comercial	800,000	9,600,000	Mensual	
10	Aguinaldo (provisión)	Aguinaldo anual ÷ 12	2,536,667	30,440,000	Mensual	
	TOTAL GASTOS FIJOS		**44,399,267**	**532,791,200**		

Appendices

Appendix A – Supporting Project Evidence

- Market Research: Microsoft Forms survey with 66 responses and Chapter 4 analysis.
- Financial model: ReFloor S.R.L. investment plan, sales forecast, purchasing model, cash flow, break-even calculation, and projected balance.
- Operational Plan: production steps, quality control, location requirements, legal environment, staffing, and safety procedures.
- Organizational Chart and Job Descriptions: Board of Directors, General Management, Finance, Commercial, HR, Production, Marketing/Social Media, and Quality/Sustainability functions.
- Brand Manual: ReFloor logo, green color palette, Arial typography, and examples of brand use.

Appendix B – Competitor Research Sources

- ECOCUR official website: https://ecocur.com.py/
- Casa Central official website: https://www.casacentralpy.com/
- Emporio de la Construcción official website: https://www.emporiodelaconstruccion.com/

Note: *Competitor evidence was checked in the project research on August 24, 2026. Prices can change; the final report therefore treats them as evidence from the research date rather than permanent prices.*